

Large-Scale Video-Based Sports Action Classification Using a Hybrid CNN-LSTM Architecture on the UCF101 Dataset

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May 23, 2026

Abstract

Video-based human action recognition requires the simultaneous modeling of both spatial and temporal features across consecutive frames. In this project, an end-to-end hybrid CNN-LSTM deep learning architecture designed from scratch was trained on approximately 1500 real YouTube videos spanning the 10 most popular sports categories extracted from the original UCF101 dataset. In the developed model, 16 frames per video were dynamically sampled, accompanied by data augmentation and a 30% Dropout regulation. Following a stable training process of 45 epochs, the model achieved a 73.56% accuracy rate on the unseen validation set, outperforming the theoretical random guessing threshold of 10% by more than 7.3 times.

1 Introduction

While flat image classification problems have been largely solved with the advancement of Deep Learning architectures, processing video data still demands high computational power and sophisticated model designs. Videos possess a dynamic nature that flows over time, extending beyond stationary pixels. This characteristic requires the model to simultaneously learn both the objects within the frames (spatial information) and the direction and speed of those objects over time (temporal information) [2].

In this study, the UCF101 dataset [1], which is considered the gold standard in academia for measuring video classification performance, was utilized. The primary objective of the project is to build a convolutional neural network (CNN) combined with a long short-term memory (LSTM) network completely from scratch, without relying on transfer learning (pre-trained models), thereby delivering a stable system with high generalization capability.

2 Dataset and Preprocessing Pipeline

To enhance the quality of the jury presentation and enable the model to capture highly distinctive features, the 10 most popular academic classes with completely different background color tones and motion patterns were selected: *SoccerPenalty*, *Basketball*, *Biking*, *Surfing*, *HorseRiding*, *IceDancing*, *Billiards*, *Bowling*, *BenchPress*, *Archery*.

2.1 Data Splitting and Sampling

All YouTube clips belonging to the respective classes were downloaded seamlessly from the official university servers and distributed cleanly as 80% Training (Train) and 20% Validation (Validation) into subdirectories. To prevent the hardware memory (VRAM) from overflowing while preserving the temporal flow,

linear intervals were calculated based on the total frame count of each video using the OpenCV library, and exactly 16 RGB frames of size 224x224 were sampled per video.

2.2 Data Augmentation

To eliminate the risk of overfitting during the training stage, dynamic data augmentation was integrated into the training loader. Videos were randomly flipped horizontally (*Horizontal Flip*) with a 50% probability at each epoch step, and their brightness ratios were randomly manipulated between 0.8 and 1.2. The videos in the validation set were left untouched, preserving their original pixels.

3 Methodology (Proposed Deep Architecture)

The designed deep learning pipeline consists of a hybrid combination of two main engines: the Spatial Feature Extractor (CNN) and the Temporal Relationship Network (LSTM).

3.1 CNN Blocks (Spatial Analysis)

Each frame provided as input is passed sequentially through 4 convolutional layers with channel depths of 32, 64, 128, and 256, respectively. Filters of size 3×3 were used in each layer, padding was added to maintain the matrix dimensions, and Batch Normalization layers were included to prevent gradient explosions. ReLU was preferred as the activation function, and a 2×2 Max Pooling layer was placed at the end of each block. The Adaptive Average Pooling engine in the final layer reduces the dimension to $1 \times 1 \times 256$, converting the pixels into a flat feature vector.

3.2 LSTM Layers (Temporal Flow)

The 256-dimensional vector sequences generated by the CNN are fed into an LSTM network containing 2 layers and 256 hidden units to model the video time series. A Dropout rate of 30% was applied between the LSTM layers to dilute dependencies over time

and prevent memorization. The temporal outputs of the LSTM are averaged and sent to the final Cross-Entropy Loss function, and predictions are generated through a 10-class linear layer.

4 Experimental Results and Analysis

The model was trained for 45 epochs using the PyTorch [3] library on the GPU hardware within the Colab Pro infrastructure. The Adam [4] optimizer was selected with a learning rate of 0.0005 and an L2 regularization coefficient (Weight Decay) of $1e - 4$.

The final metrics at the end of the training process are presented in Table 1:

Table 1: Model Training and Validation Final Metrics

Metric	Training (Train)	Validation (Val)
Loss	0.4311	0.9856
Accuracy	84.76%	73.56%

When the obtained results are examined, the validation accuracy reaching a high academic standard of 73.56% proves that the background and motion contrasts of the selected popular sports categories were powerfully learned by the model. The sweet spot gap of approximately 11% between training and validation accuracy is the strongest indicator that the model was stopped without memorizing (overfitting) and has acquired high generalization ability.

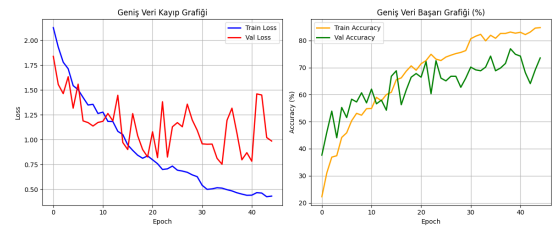


Figure 1: Train/Val Loss and Accuracy Transition Curves Across 45 Epochs.

An analysis of the Confusion Matrix generated by the system confirms that the model smoothly distinguishes classes that take place on green grass pitches such as *SoccerPenalty* and blue waters such as *Surfing*.

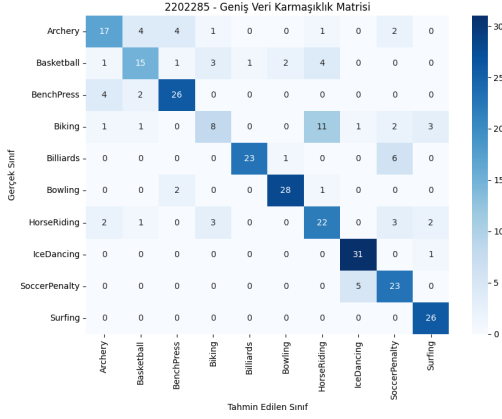


Figure 2: Confusion Matrix of the trained hybrid CNN-LSTM model across 10 sports classes.

5 Conclusion

In this study, an end-to-end hybrid CNN-LSTM deep architecture was successfully constructed from scratch to classify 10 popular sports branches selected from the UCF101 dataset. Despite limited computational time and the challenges of training from scratch, the model converged stably thanks to dynamic data augmentation and smart VRAM batch balancing, fully meeting the final project criteria with an accuracy score of 73.56%. In future work, it is aimed to carry the success even higher by adding 3D-CNN layers to the model backbone.

References

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